MILITARY SPECIFICATION

COBALT-CHROMIUM ALLOY CASTINGS (FOR BARREL TUBE LINERS)

This specification is approved for use by the Army Materials Technology Laboratory, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1.1 Scope. This specification covers one composition of cobalt-chromium alloy casting (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1190 - Minimum Guidelines for Level C Preservation, Packing and Marking
MIL-STD-1257 - Radiographic and Visual Soundness Requirements for Cobalt-Chromium Alloy Liners (For Small Arms Barrels)

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Director, U.S. Army Laboratory Command, Materials Technology Laboratory, ATTN: SLCMT-MEE, Watertown, MA 02172-0001 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D3951 - Standard Practice for Commercial Packaging
ASTM E8 - Tension Testing of Metallic Materials
ASTM E18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials
ASTM E354 - Chemical Analysis of High Temperature, Electrical, Magnetic, and Other Similar Iron, Nickel, and Cobalt Alloys

(Applications for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2 and 6.4), sample castings shall be made available to the Procuring Contracting Officer or authorized representative for approval in accordance with 4.4. The approval of the first article samples authorizes the commencement of shipment but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The first article castings shall be manufactured by the process proposed for use on production castings.

3.1.1 Process control. When specified in the contract or order (see 6.2), the supplier shall submit a quality program plan (see 6.2.2), outlining the method and procedure to be followed to ensure that castings will meet requirements. When the supplier has demonstrated that castings produced in accordance with this procurement will meet the requirements of this specification, the method and procedure will be approved and shall become a part of this specification. Changes or modifications of the approved method and procedure shall not be made without prior written approval of the Procuring Contracting Officer.

3.2 Chemical composition.

3.2.1 The chemical composition of the castings shall conform to the requirements of table I.

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TABLE I. Chemical composition, percent.

Carbon	Chromium	Molybdenum	Nickel	Iron, max.	Manganese, max.	Silicon max.	Cobalt
0.20	25.5-29.5	4.5-6.5	1.75-3.25	2.5	1.0	1.0	Remaining

3.2.2 Master heat. A certified report of the chemical composition of each master heat shall be furnished to the Procuring Contracting Officer by the contractor (see 3.4, 6.2.1 and 6.5).

3.2.3 Product analysis. When specified, a sample of metal from each lot shall be submitted to the Procuring Contracting Officer for product analysis (see 6.2).

3.3 Mechanical properties. The mechanical properties of the castings shall conform to the requirements of table II when tested in accordance with 4.5.5.

TABLE II. Mechanical properties.

Test temperature	Tensile strength (minimum), p.s.i	Elongation in 1 inch (minimum), percent	Rockwell Hardness	
			Average of all readings on a casting (or test bar)	Individual readings
Room temperature, 68° to 95°F	--	--	C24-C32	C22-C34
High temperature, 1,500°F +/-10°F	52,000	10	--	--

3.4 Production castings and finish machined liners required for visual examination.

3.4.1 Castings. Castings shall be poured directly from a master heat or from remelted master heat metal. Gates, sprues, risers, and rejected castings may be used only in the makeup of a master heat. A master heat shall not exceed 15,000 pounds.

3.4.2 Liners. Liners shall be finish machined from castings in accordance with the applicable drawing, the sampling procedure of 4.3.2.3.4.2 and the sampling requirements of MIL-STD-1257.

3.5 Dimensions. Dimensions shall be in accordance with the applicable drawings or as specified in the procurement documents (see 6.2).

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3.6 Heat treatment. Castings shall be cooled slowly in the mold to a black heat after which time cooling may be accelerated. No additional heat treatment shall be performed without approval of the Procuring Contracting Officer.

3.7 Soundness.

3.7.1 Surface.

3.7.1.1 Castings. Castings shall be free from cold shuts, fins, and other defects detrimental to fabrication or performance of parts.

3.7.1.2 Liners. Liners which have been finish machined from castings in accordance with the applicable liner drawing shall comply with the acceptance standards of MIL-STD-1257 or with standards provided in the procurement documents (see 6.2).

3.7.2 Internal. Castings shall meet the radiographic requirements of MIL-STD-1257 when examined in accordance with 4.5.2.

3.8 Identification marking. Each casting shall be identified by master heat, inspection lot, and the radiographs thereof. After radiographic examination, each acceptable casting shall be marked with the symbol X. Methods of applying identifying characters shall be as approved by the Procuring Contracting Officer. Marking materials shall have no deleterious effects on the casting or its performance.

3.9 Repair of castings. Castings shall not be repaired by plugging, peening, or welding. Unless otherwise specified in the contract or purchase order, straightening of castings shall not be permitted.

3.10 Workmanship. Castings shall be uniform in quality and condition, free from foreign materials and other injurious defects. Castings shall be well cleaned. Unless otherwise specified, metallic shot or grit shall not be used for final cleaning.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of

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ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Certification of quality compliance. A certificate of quality compliance shall be prepared for each lot of material offered for acceptance (see 6.2.2). The certificate shall include actual data of specified chemical and mechanical tests. Qualitative results of nondestructive tests and other inspections or tests shall be recorded on the certificate. The certificate shall also state that each lot has been sampled, tested, and inspected in accordance with the specification and meets all the specification requirements. The certificate shall be signed by the responsible representative of the contractor.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Sampling.

4.3.1 Lot description. A lot shall consist of 3500 pounds or less of castings of the same size and shape, poured from the same master heat, and submitted for inspection at one time.

4.3.2 Sampling for quality conformance inspection.

4.3.2.1 Sampling for visual and dimensional examination. Sampling of castings for dimensional examination, identification marking, and workmanship inspection, shall be in accordance with the provisions of MIL-STD-105, with the AQL as specified in the contract or purchase order (see 6.2).

4.3.2.2 Sampling for packaging, packing and marking. A random sample of shipping containers for the inspections of 4.5.3 shall be selected in accordance with the provisions of MIL-STD-105 and the AQL as specified in the contract or purchase order (see 6.2).

4.3.2.3 Sampling for tests.

4.3.2.3.1 For chemical analysis. At least one sample shall be taken from each lot for chemical analysis. The sample may be taken from test coupons or representative castings.

4.3.2.3.1.1 For government product analysis. When specified (see 6.2), a sample shall be taken from each lot and submitted to a designated test facility.

4.3.2.3.2 For tension tests. Four tensile specimens shall be separately cast from the same master heat and under the same condition as the lot to be

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represented. They shall be cooled at approximately the same rate as castings. Tensile test specimens shall conform to the dimensions of the largest practicable round specimen of ASTM E8.

4.3.2.3.3 For hardness tests. Sampling of castings for hardness shall be in accordance with the provisions of MIL-STD-105 with the AWL as specified in the contract or purchase order.

4.3.2.3.4 For soundness.

4.3.2.3.4.1 Castings. Each casting shall be radiographically examined for soundness.

4.3.2.3.4.2 Liners. Sample castings shall be taken from each lot in accordance with MIL-STD-105 (see 6.2) and shall be forwarded to a testing agency designated by the Procuring Contracting Officer. The castings shall be finish machined to liners and visually examined at the designated agency in accordance with MIL-STD-1257 and the AQL as specified in the contract or purchase order (see 6.2).

4.4 First article inspection. When required, the first article castings submitted in accordance with 3.1 shall be inspected for conformance to the item requirements of this specification.

4.5 Quality conformance inspection.

4.5.1 Visual and dimensional. Sample castings selected in accordance with 4.3.2.1 shall be inspected to determine conformance to the dimension requirements of 3.5, the identification marking of 3.8 and the workmanship of 3.10.

4.5.2 Soundness inspection. Sample castings and machined liners selected in accordance with 4.3.2.3.4 shall be subjected to radiographic and visual inspection in accordance with MIL-STD-1257 to determine conformance to the requirements of 3.7. Castings shall be radiographed in two positions; the second radiograph shall be taken with castings rotated 90 degrees about the longitudinal axis from the first position. The direction of radiation in each position shall be perpendicular to the longitudinal centerline of the casting and perpendicular to the plane of the radiographic film. Unless otherwise specified, film density shall be 1.4 to 2.8 for all the cross sections (see 6.2).

4.5.3 Preservation, packaging, packing and marking. The inspection shall ascertain that preservation, packaging and packing of castings and marking of containers, selected in accordance with 4.3.2.2, is in accordance with the requirements of this specification and the contract or purchase order.

4.5.4 Place of inspection. Unless otherwise specified (see 6.2), inspection and tests shall be conducted at the place of manufacture.

4.5.5 Tests.

4.5.5.1 Chemical composition. Chemical analysis shall be conducted in accordance with ASTM E354. If any sample fails to conform to the chemical requirements, the lot represented by the sample shall be rejected.

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4.5.5.2 Tension tests. Tension tests shall be conducted in accordance with ASTM E8. If a test bar fails to conform to the mechanical requirements, the lot represented by the test bar shall be subject to rejection.

4.5.5.2.1 Retests. In the event of failure of one or more representative specimens retest of additional specimens from the lot will be permitted. If one of the retest specimens fail, the lot shall be rejected with no further retesting permitted.

4.5.5.3 Hardness test. Hardness tests shall be conducted in accordance with ASTM E18. Any casting that fails to meet the hardness requirement shall be rejected and if the number of rejected exceeds the acceptance number for the sample, the lot represented by the sample shall be rejected.

4.6 Rejection.

4.6.1 Castings. Any casting that does not conform to this specification shall be rejected, and if the number of defective castings in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected. Unless otherwise specified (see 6.2), rejected lots may be offered again for Government inspection provided the contractor has removed or repaired all nonconforming castings. The Government inspector shall again select and examine samples from such rejected lots to verify compliance with this specification.

4.6.2 Liner examination. Casting lots represented by samples which fail to meet the visual soundness requirements shall be subject to rejection. Casting lots which have passed visual soundness examination will be accepted with the provision that during the life of the contract, individual liners, rejected after machining because of visible material defects, shall be replaced by sound castings at the contractor's expense.

5. PACKAGING

5.1 Preservation, packing and marking.

5.1.1 Level C. Castings shall be cleaned, preserved, packed and marked in accordance with MIL-STD-1190.

5.1.2 Commercial. Casting shall be cleaned, preserved, packed and marked in accordance with ASTM D3951.

6. NOTES

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Cobalt-chromium alloy castings covered by this specification are intended for use as small arms barrel tube liners requiring resistance to high temperature erosion.

6.2 Ordering data.

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6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation and if required, the specific issue of individual documents referenced (see 2.1).

c. Shipping instructions for first article samples (see 3.1).

d. Certified report of master heat chemical composition and product chemical analysis sample requirement (see 3.2.2 and 3.2.3).

e. Requirement for quality program plan when applicable (see 3.1.1 and 6.2.2).

f. Drawing number(s) or dimensional data (see 3.5).

g. Standards to be used in visual examination of liners if different than those of MIL-STD-1257 (see 3.7.1.2).

h. Quality of castings required for dimensional examination, identification marking, and workmanship inspection per MIL-STD-105 inspection level II, AWL 0.65% (see 4.3.2.1)

i. Quantity of shipping containers required for the packaging, packing and marking inspection per MIL-STD-105 inspection level II, AQL 4% (see 4.3.2.2).

j. Quantity of castings required for the hardness tests per MIL-STD-105, inspection level II, AWL 0.65% (see 4.3.2.3.3).

k. Quantity of castings required for liners per MIL-STD-105 inspection level S-2, identity of facility responsible for machining and visually examining machined liners per MIL-STD-1257 at an AWL of 0.65% (see 4.3.2.3.4.2).

l. When film density differing from that given in 4.5.2 is required (see 4.5.2).

m. Disposition of rejected losts (see 4.6).

n. Level C or Commercial packaging (see 5.).

6.2.2 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

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Reference Paragraph	DID Number	DID Title	Suggested Tailoring
3.1.1	DI-M-30413	Quality program plan	--
4.1.1	UDI-A-23264	Certificate of quality compliance	--

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

(Copies of data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DOD 5010.12L, Vol. 11, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094 or as directed by the contracting officer.)

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, and the number of items to be tested as specified in 4.4. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examintions, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Definition. The following term is defined for the purposes of this specification:

Master heat. A master heat is metal of a single furnace charge specifically alloyed for the requirements of this specification. The master heat may either be poured directly into castings or may be made into bars, shot, or ingots for remelting.

6.5 Subject term (key word) listing.

Castings, Cobalt-Chromium Alloy
Barrel tube liners, Small Arms

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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